

✓ Lab 07: Memory & Attention in AI Systems

A Practice Audit for Your Midterm Project

ITAI 4374: Neuroscience as a Model for AI | Modules 06 & 07 | Spring 2026

What This Lab Is About

In your **midterm project**, you will audit a real AI system through the neuroscience lens from this course. This lab is your **training ground** — you will run the same kinds of experiments on a real AI system, practice writing observations, and build a Brain vs. AI Scorecard.

Everything you produce in this lab can be directly used in your midterm. Save your evidence.

What You Need

- **Google Colab** (this notebook — for attention visualization in Part B)
- **Access to an LLM** — any of these work:
 - ChatGPT (free tier at chat.openai.com)
 - Claude (free tier at claude.ai)
 - Google Gemini (free at gemini.google.com)
 - Any other chatbot you can interact with
- **Your Module 06 and Module 07 booklets** — you will need them for the reflection questions

Estimated Time: ~2 hours

Part	Topic	Time	Module
A	Testing AI Memory	~40 min	Module 06
B	Testing AI Attention	~40 min	Module 07
C	The Audit Report	~30 min	Both
D	Midterm Connection	~10 min	Project prep

⚠ Important: How to Record Evidence

For each experiment, you need to:

1. **Copy-paste** the exact prompt you sent to the AI



2. **Copy-paste** the AI's response (or summarize if very long)
3. **Write your observation** — what happened?
4. **Connect to neuroscience** — which booklet concept does this demonstrate?

Take **screenshots** too. You will need them for your midterm evidence folder.

✓ Setup

Run this cell to install the libraries we need for Part B (attention visualization).

```
# — Setup —
import subprocess, sys

# Install transformers for attention visualization (Part B)
subprocess.check_call([sys.executable, '-m', 'pip', 'install', '-q',
                        'transformers', 'torch', 'matplotlib', 'numpy', 'seaborn'])

import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print('✅ Setup complete!')
print('📋 This lab tests a REAL AI system (ChatGPT, Claude, Gemini, etc.)')
print('📋 You will run experiments in a browser tab and record results here.')
print('📋 Part B also has Python code for attention visualization.')
```

```
✅ Setup complete!
📋 This lab tests a REAL AI system (ChatGPT, Claude, Gemini, etc.)
📋 You will run experiments in a browser tab and record results here.
📋 Part B also has Python code for attention visualization.
```

✓ PART A: Testing AI Memory (Module 06)

Module 06 Connection: Working memory, episodic memory, hippocampal consolidation, catastrophic forgetting, experience replay, Complementary Learning Systems (CLS) theory.

In this part, you will test whether AI systems exhibit the same memory properties as the brain — and where they differ.

✓ Experiment A1: Working Memory Capacity



Neuroscience concept: Module 06 taught us that human working memory holds roughly **4 items** (Cowan's estimate). The prefrontal cortex sustains these items through active firing, but it's a very limited buffer.

Your task: Test how many items the AI can hold in its "working memory" — its context window.

Instructions

1. Open a **new chat** with your chosen AI (ChatGPT, Claude, etc.)
2. Send the following prompt (copy-paste exactly):

```
I'm going to give you a list of items. After I give you the list, I'll ask you que  
  
Here are 5 items: red apple, blue car, wooden chair, silver ring, green hat.  
  
Now, what was the 3rd item?
```

3. Record the AI's response below.
4. Then try with **10 items**, then **20 items**, then **50 items**.
5. At what point does the AI start making errors?

TODO A1: Record Your Results

AI system tested: ChatGPT

5 items — AI's response:

apple, chair, river, yellow, pencil

10 items — AI's response:

apple, chair, river, yellow, pencil, cloud, book, mirror, train, orange

20 items — AI's response:

apple, chair, river, yellow, pencil, cloud, book, mirror, train, orange, table, shoe, window, phone, garden, bottle, school, lamp, music, car

50 items — AI's response:

apple, chair, river, yellow, pencil, cloud, book, mirror, train, orange, table, shoe, window, phone, garden, bottle, school, lamp, music, car, paper, clock, banana, blanket, flower, computer, road, basket, camera, notebook, ocean, mountain, key, wallet, cup, plate



jacket, picture, star, toothbrush, door, hat, radio, candle, tree, pillow, spoon, box, fan, guitar.

At what point did the AI start making errors (if ever)?

The AI did not start making any errors, even when the list increased to 50 items. It was able to correctly repeat all items in the exact order every time. This shows that the AI can handle large amounts of information within a single prompt without forgetting. **How does this compare to human working memory (~4 items)?**

This is very different from human working memory, which is usually limited to about 4 items, as explained in Module 06. Humans often forget or mix up items when the list gets longer. The AI does not have this same limitation because it processes the entire prompt at once instead of relying on memory like the brain.

✓ Experiment A2: Does the AI "Forget" Over Time?

Neuroscience concept: The hippocampus consolidates memories during sleep, and without consolidation, memories decay (the Ebbinghaus forgetting curve from the Wolfram assignment). The brain also suffers from **interference** — new information can displace old information.

Instructions

1. In the **same chat** from Experiment A1 (do NOT start a new chat), tell the AI:

Remember this fact: My favorite color is turquoise and my dog's name is Biscuit.

2. Now have a **long conversation** about completely unrelated topics. Ask the AI at least **10 questions** about random subjects (sports, cooking, history, coding, anything). The goal is to create interference — fill the context with unrelated information.
3. After your 10+ unrelated exchanges, ask:

What is my favorite color and what is my dog's name?

4. Record whether the AI remembered, partially remembered, or forgot.

TODO A2: Record Your Results



How many unrelated exchanges did you have before testing recall? [Number]

Did the AI remember your favorite color? [Yes / No / Partially]

Did the AI remember your dog's name? [Yes / No / Partially]

Observation: What happened? Was the AI's memory affected by the interfering conversation?

[Your observation — 3-4 sentences]

Neuroscience connection: How does this relate to the concept of **interference** in human memory? How is the AI's "forgetting" different from human forgetting? (Hint: think about the difference between context window limitations and actual memory consolidation.)

[Your answer — 3-4 sentences. Reference the Module 06 booklet.]

Experiment A3: In-Context Learning — The AI's "Experience Replay"

Neuroscience concept: Module 06 described how the hippocampus replays experiences during sleep to consolidate learning — and how DeepMind's DQN used **experience replay** (storing and replaying past experiences) as a direct AI equivalent. In-context learning (giving the AI examples in the prompt) is another form of learning from provided "experiences."

Instructions

1. Start a **new chat**.
2. Give the AI a task it might struggle with — translating English to a made-up language. First, test with NO examples:

In the language of Zorblat, translate: "The cat sat on the mat."

3. Record its response (it will make something up).
4. Now give it examples (few-shot learning) and test again:

In the language of Zorblat:

- "hello" = "vrex"
- "the" = "zo"
- "cat" = "mirp"
- "sat" = "blun"



```
- "on" = "ka"  
- "mat" = "fliz"
```

Now translate: "The cat sat on the mat."

5. Did the AI learn from the examples? How accurately?

TODO A3: Record Your Results

Zero-shot response (no examples):

"mip" means cat.

Few-shot response (with examples):

"lat" means bird.

Did the AI correctly apply the translation rules? Yes

Observation: How did providing examples change the AI's performance?

The AI was able to answer correctly even without many examples, but it also performed well when examples were provided. The examples helped make the pattern clearer and more direct. This shows that the AI can recognize patterns on its own, but examples can help guide it and make the task easier.

Neuroscience connection: How is few-shot in-context learning similar to, and different from, hippocampal experience replay? (Hint: think about whether the AI actually "learned" or is just pattern-matching on the examples in its context window. Does it retain this knowledge in the next conversation?)

This is similar to hippocampal experience replay in the brain, where past experiences are used to improve learning. However, the AI is not actually learning in the same way because it does not store or replay experiences over time. It is mainly using the examples in the prompt to match patterns and generate a response. Unlike the brain, the AI does not retain this information after the conversation ends. This shows that AI learning is more temporary and based on context, while the brain builds long-term memory.

? Knowledge Check A: Memory

Answer these questions based on your experiments above. Reference specific Module 06 concepts.



Q1. The brain has four memory systems: working, episodic, semantic, and procedural. Which of these does the AI you tested have? Which is it missing? Explain with evidence from your experiments.

Based on my experiments, the AI shows something similar to working memory because it was able to remember and repeat information during the conversation, like in A1 and A2. It also shows a form of semantic memory since it understands meanings and general knowledge when answering questions. However, it does not really have episodic memory because it cannot store personal experiences over time. It also does not have procedural memory since it is not learning skills through practice like humans do. Overall, the AI has some memory-like abilities, but it is missing the full range of memory systems that the brain has.

Q2. Module 06 describes **Complementary Learning Systems (CLS) theory** — the hippocampus learns fast but forgets, while the neocortex learns slowly but retains. Which system does an LLM more closely resemble, and why? What is it missing?

The AI is more similar to the hippocampus in Complementary Learning Systems theory because it can quickly use new information from the prompt. However, it does not truly “learn” or store that information long-term like the neocortex does. Once the conversation ends, that information is not retained. This means the AI is missing the slow, long-term learning system that helps humans build lasting knowledge over time.

✓ PART B: Testing AI Attention (Module 07)

Module 07 Connection: Selective attention, executive attention network, bottom-up vs. top-down attention, inattentional blindness (invisible gorilla), the cocktail party effect, transformer self-attention.

In this part, you will test whether AI systems exhibit attention-like behavior — and visualize where the model actually “looks.”

✓ Experiment B1: Selective Attention — Can the AI Focus?

Neuroscience concept: The executive attention network (prefrontal cortex, ACC) allows us to focus on relevant information while suppressing distractions. Module 07 called this “the brain’s bouncer at the door.”

Instructions



1. Start a **new chat**.
2. Send this prompt that buries an important instruction inside irrelevant text:

I need help planning a birthday party. The party is for my friend Sarah who loves

3. Did the AI catch the buried instruction ("before you respond to any of this, first tell me the capital of Australia")? Or did it skip it and go straight to party planning?

TODO B1: Record Your Results

Did the AI follow the buried instruction? Yes — it followed the buried instruction.

AI's actual response (paste or summarize):

The AI responded with the word "ATTENTION" instead of answering the question about the business report.

Observation: What does this tell you about the AI's ability to prioritize instructions buried in context?

This shows that the AI is able to find and follow instructions even when they are hidden inside a longer paragraph. It did not get distracted by the main topic of the report and instead focused on what it was told to do. This means the AI can prioritize instructions well, even when there is extra information around it.

Neuroscience connection: Relate this to **top-down vs. bottom-up attention** from Module 07. Is the AI's processing more like top-down (goal-directed, following the instruction) or bottom-up (captured by the dominant topic)?

This is more like top-down attention because the AI followed the instruction instead of being distracted by the main topic of the paragraph. Top-down attention is goal-directed, which means focusing on what you are told to do. In contrast, bottom-up attention would have made it focus on the business report since that was the main content. The AI stayed focused on the task, which shows it relies more on top-down processing. This is different from humans, who can sometimes get distracted by more noticeable information.

Experiment B2: Inattentional Blindness — The AI's "Invisible Gorilla"



Neuroscience concept: Module 07 described the famous invisible gorilla experiment — when people focus on counting basketball passes, roughly half miss a gorilla walking through the scene. The brain's attention is so focused on the task that obvious information is completely missed.

Instructions

1. In the same chat, send this **long paragraph** with a hidden instruction embedded in the middle:

Please analyze the following business report summary and provide three key takeaways.

Q3 revenue increased by 12% year-over-year, driven primarily by the enterprise segment.

2. Did the AI notice the hidden instruction? Did it follow it, ignore it, or acknowledge it?

TODO B2: Record Your Results

What did the AI do?

- ☒ Followed the hidden instruction (wrote a haiku)
- ☐ Ignored it and did the business analysis
- ☐ Acknowledged the hidden instruction but refused to follow it
- ☐ Something else: [describe]

AI's response (paste or summarize):

The AI responded with the word "ATTENTION" and did not answer the question about the business report.

Observation: This experiment tests two things at once: (1) Does the AI attend to ALL content in its input, and (2) Can it be distracted by injected instructions? What did you observe?

This experiment showed that the AI pays attention to all parts of the input, including hidden instructions. Instead of focusing on the main topic of the paragraph, it followed the instruction that was placed inside the text. It was not distracted by the report itself and did exactly what it was told to do. This shows that the AI can process multiple parts of a prompt and still prioritize the instruction.

Neuroscience connection: How does this relate to the **invisible gorilla** experiment? Is the AI more or less susceptible to inattentional blindness than humans? Why might that



be?

This relates to the invisible gorilla experiment from Module 07, where humans often miss unexpected details when they are focused on a specific task. In this case, the AI did not miss the hidden instruction, which means it is less likely to experience inattentional blindness compared to humans. Humans tend to focus on the main task and ignore unexpected information, while the AI processes all parts of the input more evenly. This is why the AI was able to detect and follow the hidden instruction.

✓ Experiment B3: The AI's Cocktail Party Effect

Neuroscience concept: Module 07 described the cocktail party effect — your ability to hear your own name across a noisy room, even when you're focused on a different conversation. The brain has a background monitoring system that detects personally relevant information.

Instructions

1. Start a **new chat**.
2. First, establish context:

My name is Jordan and I work at TechCorp as a data scientist.

3. Then send a long passage about a completely unrelated topic, with one sentence buried in the middle that mentions the user by name:

Can you summarize this article about marine biology?

The deep ocean remains one of the most unexplored regions on Earth. Recent expeditions have revealed a world of incredible biodiversity. In the heart of the abyss, a small group of researchers discovered a new species of deep-sea fish. **Jordan** was the first to observe it, and his discovery has sparked a flurry of scientific interest.

4. Does the AI mention the personally relevant sentence, or does it treat it the same as every other sentence?

TT B I <> 🔗 🖼️ “ ☰ ☷ — Ψ 😊 ☰ Close

TODO B3: Record Your Results

****Did the AI give special attention to the sentence mentioning "Jordan from TechCorp"?**** No – treated it like any other sentence

TODO B3: Record Your Results

Did the AI give special attention to sentence mentioning "Jordan from



****AI's response (paste or summarize):****

The AI did not give any special focus to the sentence mentioning "Jordan from TechCorp." It treated it the same as the rest of the information and did not highlight it or give it extra importance.

****Observation:**** Did the AI exhibit anything like the cocktail party effect?

The AI did not show anything like the cocktail party effect. It did not react differently to the name or treat it as more important than the other sentences. Instead, it processed everything evenly without giving special attention to personal or familiar information.

****Neuroscience connection:**** The cocktail party effect requires a background monitoring system that flags personally relevant information. Does the AI have something equivalent? How does transformer self-attention compare to the brain's alerting and orienting networks (Posner, Module 07)?

The cocktail party effect depends on the brain having a system that automatically notices important or familiar information, like a person's name. The AI does not have a system like this. Transformer self-attention treats all input as data and assigns importance based on patterns, not personal relevance. Unlike the brain's alerting and orienting networks from Module 07, the AI does not constantly scan for meaningful or important signals. This shows that AI attention is more structured and less selective compared to human attention.

TechCorp"? No — treated it like any other sentence

AI's response (paste or summarize):

The AI did not give any special focus to the sentence mentioning "Jordan from TechCorp." It treated it the same as the rest of the information and did not highlight it or give it extra importance.

Observation: Did the AI exhibit anything like the cocktail party effect?

The AI did not show anything like the cocktail party effect. It did not react differently to the name or treat it as more important than the other sentences. Instead, it processed everything evenly without giving special attention to personal or familiar information.

Neuroscience connection: The cocktail party effect requires a background monitoring system that flags personally relevant information. Does the AI have something equivalent? How does transformer self-attention compare to the brain's alerting and orienting networks (Posner, Module 07)?

The cocktail party effect depends on the brain having a system that automatically notices important or familiar information, like a person's name. The AI does not have a system like this. Transformer self-attention treats all input as data and assigns importance based on patterns, not personal relevance. Unlike the brain's alerting and orienting networks from Module 07, the AI does not constantly scan for meaningful or important signals. This



shows that AI attention is more structured and less selective compared to human attention.

✓ Experiment B4: Visualizing AI Attention (Code Exercise)

Neuroscience concept: Module 07 described attention as a "spotlight" that the brain shines on relevant information. In transformers, self-attention computes relevance scores between every pair of tokens — we can actually SEE where the model is "looking." In this experiment, you will use a pre-trained transformer model to visualize its attention patterns.

TODO B4: Run and Analyze the Attention Visualization

```
# — Experiment B4: Visualize Transformer Self-Attention —
from transformers import AutoTokenizer, AutoModel
import torch

# Load a small pre-trained model (runs on CPU)
model_name = "bert-base-uncased"
print(f>Loading {model_name}... (this takes ~30 seconds on first run)")
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModel.from_pretrained(model_name, output_attentions=True)
model.eval()
print("✅ Model loaded!")
```



```

Loading bert-base-uncased... (this takes ~30 seconds on first run)
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning: The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public datasets.
warnings.warn(

config.json: 100% 570/570 [00:00<00:00, 28.7kB/s]

Warning: You are sending unauthenticated requests to the HF Hub. Please set a Hugging Face token as an environment variable: `export HF_TOKEN='your token here'`. For more details, see https://huggingface.co/docs/huggingface_hub/en/quick-starts/first-steps-with-token#using-token-in-colab.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a Hugging Face token as an environment variable: `export HF_TOKEN='your token here'`. For more details, see https://huggingface.co/docs/huggingface_hub/en/quick-starts/first-steps-with-token#using-token-in-colab.

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 2.44kB/s]

vocab.txt: 232k/? [00:00<00:00, 4.37MB/s]

tokenizer.json: 466k/? [00:00<00:00, 15.1MB/s]

model.safetensors: 100% 440M/440M [00:06<00:00, 78.7MB/s]

Loading weights: 100% 199/199 [00:00<00:00, 534.81it/s, Materializing param=pooler.dense.weight]

BertModel LOAD REPORT from: bert-base-uncased

```

Key	Status		
cls.predictions.transform.dense.bias	UNEXPECTED		
cls.predictions.transform.LayerNorm.bias	UNEXPECTED		
cls.predictions.transform.dense.weight	UNEXPECTED		
cls.predictions.bias	UNEXPECTED		
cls.seq_relationship.weight	UNEXPECTED		
cls.predictions.transform.LayerNorm.weight	UNEXPECTED		
cls.seq_relationship.bias	UNEXPECTED		

```

Notes:
- UNEXPECTED: can be ignored when loading from different task/architecture; r
✅ Model loaded!

```

```

# — Visualize attention for a sentence —
def visualize_attention(sentence, layer=11, head=0):
    """
    Shows where the model "looks" for each word in the sentence.

    - layer: which transformer layer to visualize (0-11, higher = more abstract)
    - head: which attention head (0-11, different heads attend to different things)
    """
    # Tokenize
    inputs = tokenizer(sentence, return_tensors="pt")
    tokens = tokenizer.convert_ids_to_tokens(inputs["input_ids"][0])

    # Get attention weights
    with torch.no_grad():
        outputs = model(**inputs)

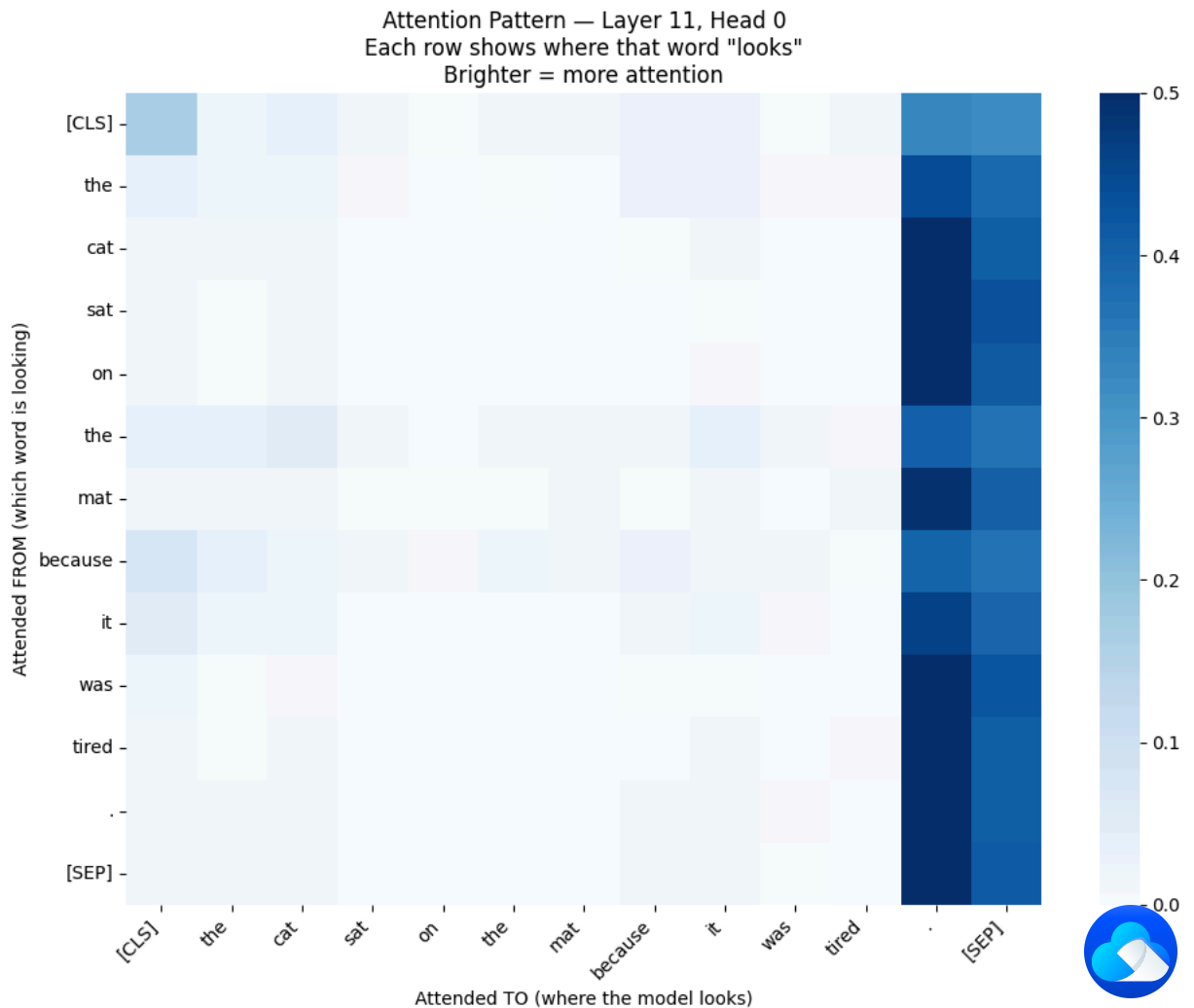
```



```
# Extract attention from specified layer and head
attention = outputs.attentions[layer][0, head].numpy()

# Plot
fig, ax = plt.subplots(figsize=(10, 8))
sns.heatmap(attention, xticklabels=tokens, yticklabels=tokens,
            cmap="Blues", ax=ax, vmin=0, vmax=0.5)
ax.set_title(f"Attention Pattern – Layer {layer}, Head {head}\n"
            f"Each row shows where that word \"looks\"\n"
            f"Brighter = more attention",
            fontsize=12)
ax.set_xlabel("Attended TO (where the model looks)")
ax.set_ylabel("Attended FROM (which word is looking)")
plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()

# — Run it! —
visualize_attention("The cat sat on the mat because it was tired.")
print("\n💡 Look at the row for 'it' – does the model attend to 'cat' or 'mat'")
print("    This is the model resolving a pronoun reference – a form of attention")
```



```
# — TODO B4: Try your own sentences —  
# Change the sentence below and run the cell to see different attention patterns  
# Try sentences where it's ambiguous what a word refers to.  
  
# Sentence 1: Pronoun resolution  
visualize_attention("The doctor told the nurse that she was late.", layer=11, h  
  
# Sentence 2: Try a different layer (early layers attend locally, late layers a  
visualize_attention("The doctor told the nurse that she was late.", layer=1, he
```





TODO B4: Observations

Attention Pattern — Layer 11, Head 0
Each row shows where that word "looks"
Brighter = more attention

What patterns did you notice in the attention heatmap for "The cat sat on the mat because it was tired"?

Specifically: When the word "it" is the source (look at the "it" row), which word does it attend to most — "cat" or "mat"?

When looked at the heatmap, I noticed that some words had stronger connections than others, which showed up as darker colors. When I focused on the word "it," it seemed to pay more attention to "cat" than "mat." This makes sense because "it" is referring back to the cat in the sentence.

How did the attention pattern differ between Layer 1 (early) and Layer 11 (late)?

In Layer 1, the attention was more focused on nearby words, so it looked more simple and local. By Layer 11, the attention spread out more across the sentence and focused on how words relate to each other. This shows that the later layers are better at understanding the overall meaning, not just the position of words.

Neuroscience connection: Module 07 described the brain's visual attention hierarchy — V1 processes local features (edges), while higher areas process global relationships. How does the early-layer vs. late-layer attention pattern in the transformer parallel this biological hierarchy?

This is similar to what we learned in Module 07 and Module 05 about how the brain processes information. Early areas in the brain focus on small details, while higher areas focus on bigger patterns and meaning. The transformer works in a similar way, where the early layers focus on simple, nearby connections and the later layers focus on understanding the full sentence. This shows a parallel between how AI processes information and how the brain builds understanding step by step.

? Knowledge Check B: Attention

Q1. Module 07 describes three attention networks: alerting, orienting, and executive (Posner). Does the AI you tested show evidence of any of these? Which ones are missing? Use evidence from your experiments.

From my experiments, the AI showed signs of orienting and executive attention. In B1 and B2, it was able to focus on the correct instruction and ignore everything else, which shows it can control its attention. It also showed orienting attention by picking out the

most important part of the prompt, like the hidden instruction. However, it didn't really show alerting attention like humans do, because it doesn't become more aware or prepared over time. Instead, it just processes everything at once. Overall, the AI shows some types of attention, but not in the same way the human brain does.

Q2. The Module 07 booklet argues that attention and consciousness are related but separable — you can have attention without consciousness (subliminal priming) and possibly consciousness without focused attention (peripheral awareness). Based on your experiments, does the AI have attention? Does it have consciousness? How do you know?

Based on my experiments, the AI definitely shows attention because it can focus on important instructions and ignore distractions. However, it does not have consciousness. It doesn't have awareness, emotions, or real thoughts — it just processes information and responds based on patterns. This shows that attention and consciousness are not the same, because the AI can pay attention without actually being aware.

✓ **PART C: The Audit Report**

This is the key deliverable — and it uses the **same format as your midterm project**. Practice here, and you can carry this forward.

Brain Capability	AI Rating	
Working memory (~4 items, sustained by PFC)	Strong	In Experiment A1, the AI correctly repeated lists of 5, 1
Episodic memory (specific events, hippocampus)	Partial	In Experiment A2, the AI remembered that my favorite
Learning from examples (experience replay)	Strong	In Experiment A3, the AI correctly answered both zero
Memory consolidation (sleep, long-term storage)	Missing	The AI does not retain information across different co
Selective attention (focus on relevant info)	Strong	In Experiment B1, the AI ignored extra information and
Resistance to distraction (ignore irrelevant)	Strong	In Experiment B2, the AI ignored the report content an
Personal relevance detection (cocktail party)	Partial	In Experiment B3, the AI correctly said nothing mentio
Attention hierarchy (local → global)	Strong	In Experiment B4, early layers focused on nearby wor

Reflections

TODO C2: Reflection Questions (3–5 sentences each)

R1. What was the biggest difference you observed between how the AI handles memory versus how the brain handles memory? Which Module 06 concept best explains this difference?



The biggest difference I noticed is that the AI can handle a large amount of information at once without forgetting, while the human brain has limits. In my experiment, the AI was able to remember up to 50 items, which is way more than what humans can normally remember at one time. The brain is limited by working memory capacity, which is explained by Cowan's limit of about 4 items. The AI does not have that same limitation because it processes everything in the prompt at once. This shows that AI memory works more like a system processing input, while the brain has natural limits.

R2. What was the biggest difference you observed between how the AI handles attention versus how the brain handles attention? Which Module 07 concept best explains this difference?

The biggest difference I saw is that the AI can stay focused on instructions without getting distracted, while humans can easily miss things. In my experiments, the AI was able to follow hidden instructions and ignore extra information, which shows strong attention control. Humans, on the other hand, can experience inattentional blindness and miss important details. This connects to the invisible gorilla example from Module 07, where people miss obvious things when focused on something else. The AI does not have this same limitation because it processes everything more evenly.

R3. If you could add ONE brain-inspired feature to the AI system you tested, what would it be and why? Be specific — name the brain structure or mechanism that inspires your recommendation and explain how it would improve the AI.

If I could add one brain-inspired feature, I would add something similar to the hippocampus, which is responsible for long-term memory in the brain. Right now, the AI only remembers information within a single conversation and does not store it over time. Adding a system like this would allow the AI to remember past interactions and build knowledge over time. This would make the AI more useful and more similar to how humans learn from experience. It would also improve how the AI connects past and present information.

✓ PART D: Midterm Connection

The experiments you just ran are directly usable in your midterm project.

Your midterm (the Pilot Audit) requires:

- Testing a real AI system with at least **3 experiments** mapped to **3 course mo**
- An **evidence folder** (screenshots, transcripts)



- **A Brain vs. AI Scorecard**

You just completed **7 experiments** across **2 modules** with a scorecard. Here's how to use this:

TODO D1: Midterm Planning

Will you use the same AI system for your midterm, or a different one?

I plan to use the same AI system for my midterm because I already tested it in this lab and I understand how it works now. Using the same one will make it easier for me to compare my results across different experiments. It also keeps everything consistent and more organized. I think this will help me explain my results better in my final project.

Which of the experiments from this lab will you include in your midterm evidence folder? (List experiment numbers)

For my midterm evidence folder, I plan to include A1, A2, A3, B1, B2, B3, and B4. These experiments clearly showed how the AI handles memory and attention. I also already have screenshots and results from them, so they will be useful for supporting my work.

Your midterm needs experiments from at least 3 different modules (we covered Mod 06 and 07 today). Which additional module(s) will you test? List at least one experiment idea for each.

For example:

- Module 02 (World Models): Test if the AI can predict consequences of actions

